Organisation

Break down food

1 Organisation & digestion

Cells → tissues → organs → systems.

Digestion: large insoluble molecules → small soluble molecules; absorption into blood.

2 Enzymes & tests

Amylase → sugars; protease → amino acids; lipase → glycerol/fatty acids.

Bile neutralises acid and emulsifies fat. Enzymes have specific active sites.

Food tests identify nutrients; pH investigation uses iodine and **rate = 1/time.**

Transport & protect

3 Heart, blood & lungs

Right heart → lungs; left heart → body. Valves prevent backflow.

Arteries leave; veins return; capillaries exchange. Red cells carry O_2; white cells defend; platelets clot.

Alveoli: large area, thin barrier; blood/ventilation maintain gradients.

4 Health & disease

Coronary deposits reduce oxygen supply; stents open arteries, statins lower cholesterol.

Risk factors raise probability. Correlation alone is not causation.

Benign tumours stay local; malignant cells invade/spread.

Plant systems

5 Leaf structure

Palisade chloroplasts absorb light; spongy air spaces allow diffusion.

Epidermis/cuticle protect; guard cells control stomata. Meristems divide at growing tips.

6 Transport in plants

Root hairs: water by osmosis, minerals by active transport.

Xylem carries water/minerals; phloem translocates sugars.

Transpiration rises with heat/wind/light, falls with humidity. Potometers measure uptake.